

PALMAS, Brazil

WMO: 830650

Lat: 10.283S

Lon: 48.350W

Elev: 236

StdP: 98.52

Time Zone: -3.00 (W03)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.0	19.9	7.2	6.5	32.4	8.5	7.1	30.9	7.2	26.3	6.1	26.8	2.1	100	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	12.0	37.1	20.9	36.2	21.0	35.8	21.0	26.0	30.3	25.7	30.0	25.4	29.8	3.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.7	27.6	24.3	19.9	27.2	24.1	19.6	27.1	82.0	30.6	80.4	30.1	79.2	29.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.8	5.1	17.2	38.7	1.0	1.3	16.5	39.6	15.9	40.4	15.4	41.1	14.6	42.0		
			13.7	26.6	0.9	0.6	13.0	27.1	12.5	27.4	11.9	27.7	11.3	28.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.8	26.7	27.0	27.4	27.5	27.0	27.0	28.5	29.9	28.8	27.6	27.2
	DBStd	1.60	1.32	1.51	1.30	1.07	1.02	0.89	1.00	1.23	1.32	1.82	1.51	1.45
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6438	520	469	528	522	544	511	527	573	598	584	529	533
	CDD18.3	3396	262	236	270	272	286	261	268	314	348	326	279	274
	CDH23.3	34782	2132	1914	2326	2464	2859	2732	2999	3912	4591	3698	2699	2456
	CDH26.7	16005	734	673	806	943	1303	1378	1592	2191	2586	1813	1062	926
Wind	WSAvg	2.3	2.0	2.0	1.9	1.9	2.1	2.5	2.7	2.7	2.6	2.3	2.1	2.1
Precipitation	PrecAvg	1719	301	271	270	160	55	3	1	6	48	127	208	269
	PrecMax	2106	599	396	502	299	181	25	8	29	123	254	314	513
	PrecMin	1271	141	76	149	54	4	0	0	0	5	53	102	122
	PrecStd	228	127	81	89	58	35	6	2	5	36	51	53	101
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.8	34.1	34.8	34.2	34.9	35.2	37.1	38.1	38.0	36.1	35.2
		MCWB	23.7	23.7	24.8	23.4	23.1	22.2	20.3	20.3	20.3	21.1	23.2	23.6
	2%	DB	33.1	33.8	32.9	33.8	34.0	34.0	34.8	36.2	37.2	36.9	34.2	33.9
		MCWB	24.3	24.2	24.7	23.9	23.0	21.5	20.3	20.1	20.6	21.6	23.8	24.0
	5%	DB	31.9	31.9	32.0	32.8	33.1	33.2	34.0	35.2	36.8	35.9	33.1	32.3
		MCWB	24.2	24.2	24.6	24.3	23.2	21.3	20.2	20.1	20.6	21.8	24.0	24.3
	10%	DB	30.5	30.2	30.9	31.2	32.2	32.9	33.2	34.8	35.9	34.1	31.9	31.1
		MCWB	24.3	24.3	24.6	24.4	23.2	21.3	20.1	20.1	20.6	22.5	24.1	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.1	26.5	26.2	25.9	25.0	23.3	23.5	24.7	25.6	26.1	26.2
		MCDB	30.3	30.4	30.4	30.1	30.4	30.1	30.2	31.2	31.2	31.0	30.8	31.0
	2%	WB	25.5	25.5	26.0	25.7	25.3	24.1	22.4	22.6	24.2	25.0	25.5	25.6
		MCDB	29.8	29.9	29.9	29.7	30.2	30.0	30.6	31.1	30.6	30.8	30.4	30.4
	5%	WB	25.1	25.1	25.6	25.4	24.8	23.4	21.6	21.8	23.7	24.5	25.0	25.1
		MCDB	29.3	29.4	29.5	29.5	29.8	29.7	30.6	31.1	30.3	30.1	30.1	29.8
	10%	WB	24.6	24.6	25.1	25.1	24.4	22.8	21.0	21.1	23.2	24.1	24.6	24.6
		MCDB	28.8	28.8	29.0	29.1	29.4	29.3	30.4	31.6	30.2	29.5	29.5	29.2
Mean Daily Temperature Range	5% DB	MDBR	8.1	7.9	7.9	8.5	10.0	12.3	13.6	13.5	12.0	10.1	8.5	8.3
		MCDBR	10.0	10.0	9.5	10.4	11.2	12.9	14.3	14.0	13.4	12.5	10.5	10.2
		MCWBR	3.5	3.6	3.4	3.6	3.6	4.2	4.5	4.8	4.9	4.0	3.4	3.5
	5% WB	MCDBR	8.6	8.6	8.4	8.7	9.7	11.8	13.9	13.6	11.4	9.9	9.2	9.2
		MCWBR	3.6	3.6	3.5	3.4	3.5	4.4	5.0	5.1	4.2	3.4	3.4	3.6
Clear-Sky Solar Irradiance	taub	0.406	0.405	0.404	0.401	0.389	0.353	0.351	0.402	0.517	0.521	0.459	0.437	
	taud	2.456	2.465	2.476	2.464	2.491	2.574	2.550	2.365	2.036	2.083	2.281	2.352	
	Ebn at Noon	936	932	917	889	870	889	899	875	805	820	881	905	
	Edh at Noon	120	119	115	111	102	92	95	121	176	172	142	133	
All-Sky Solar Radiation	RadAvg	5.24	5.19	5.06	5.20	5.25	5.47	5.79	6.25	6.04	5.53	5.23	5.23	
	RadStd	0.49	0.60	0.25	0.34	0.28	0.16	0.15	0.27	0.48	0.42	0.37	0.35	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page